=

Program Developed By:

Name: JANA SHRAVIN S

Register Number: 212224243003

=

In [3]:

```
import cv2
import matplotlib.pyplot as plt
import numpy as np
image1=cv2.imread("shravin.jpeg")
image2=cv2.cvtColor(image1,cv2.COLOR_BGR2RGB)
kernel=np.ones((11,11),np.float32)/169
image3=cv2.filter2D(image2,-1,kernel)
plt.figure(figsize=(9,9))
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(image3)
plt.title("Average Filter Image")
plt.axis("off")
plt.show()
```

Original Image　　　　Average Filter Image

In [4]:

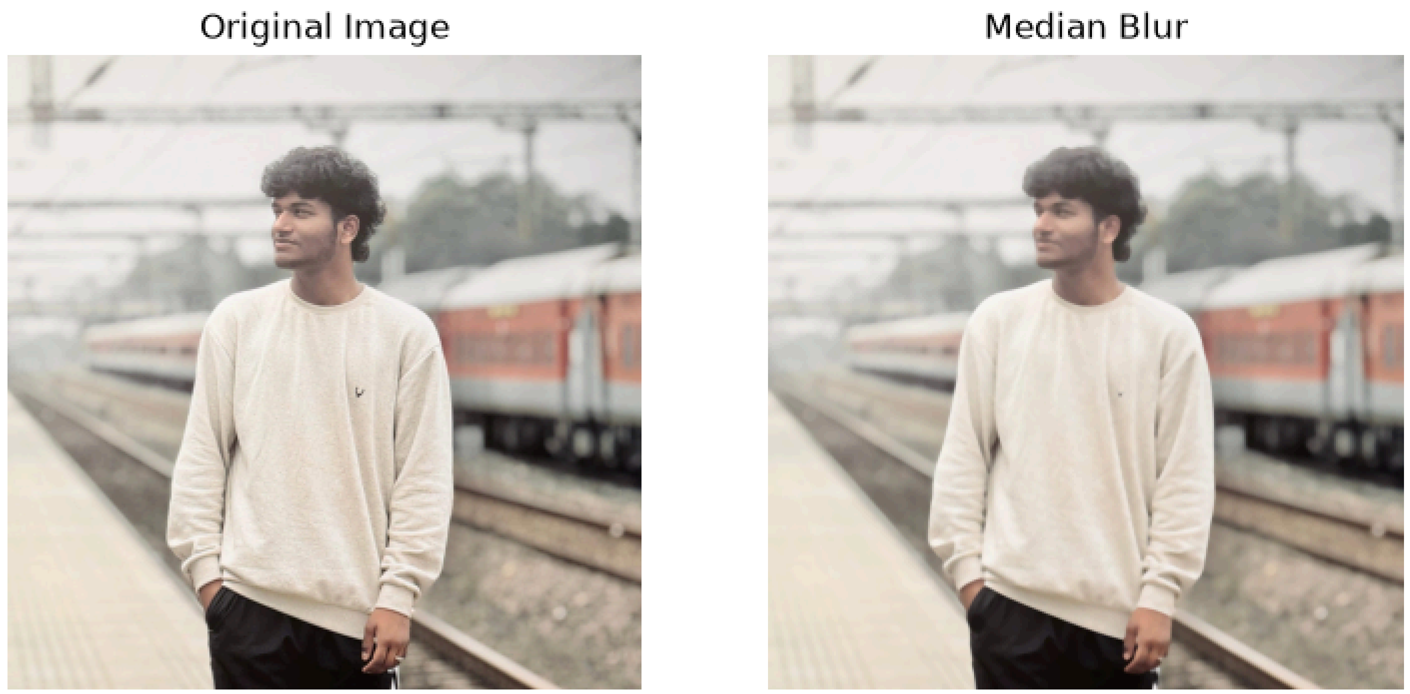
```
kernel1=np.array([[1,2,1],[2,4,2],[1,2,1]])/16
image2=cv2.cvtColor(image1,cv2.COLOR_BGR2RGB)
image3=cv2.filter2D(image2,-1,kernel1)
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(image3)
plt.title("Weighted Average Filter Image")
plt.axis("off")
plt.show()
```

In [5]:

```
gaussian_blur=cv2.GaussianBlur(image2,(33,33),0,0)
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(gaussian_blur)
plt.title("Gaussian Blur")
plt.axis("off")
plt.show()
```

In [6]:

```
median=cv2.medianBlur(image2,13)
plt.figure(figsize=(9,9))
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(median)
plt.title("Median Blur")
plt.axis("off")
plt.show()
```



In [7]:

```python
kernel2=np.array([[-1,-1,-1],[2,-2,1],[2,1,-1]])
image3=cv2.filter2D(image2,-1,kernel2)
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(image3)
plt.title("Laplacian Kernel")
plt.axis("off")
plt.show()
```

In [8]:

```python
laplacian=cv2.Laplacian(image2,cv2.CV_64F)
plt.subplot(1,2,1)
plt.imshow(image2)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(laplacian)
plt.title("Laplacian Operator")
plt.axis("off")
plt.show()
```

```
Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).
Got range [-174.0..256.0].
```

Original Image

Laplacian Operator